

Sahana Defence Limited

Frontend Mobile Responsiveness & QA CSS Audit Report

Target: Live Website (WordPress + React Frontend)
Date: September 2026
Version: Production Release 2.0

Developer Context: This document outlines all frontend sections, components, and CSS rules refactored during comprehensive mobile UX audits. The live website architecture features a **WordPress Backend** serving the compiled **React / Tailwind CSS Frontend**. Use this reference to verify parity across your live production theme and CSS bundles.

1 Core Architectural CSS Rules (theme.css)

These foundational classes in `src/styles/theme.css` regulate responsiveness and touch interactions globally:

Property / Utility	Implementation	Purpose & Benefit
Dynamic Viewport Height	<code>min-h-[100dvh]</code> (replaced fixed <code>h-screen</code>)	Eliminates UI jumping and cutoffs on iOS Safari and Chrome Android caused by dynamic URL address bar appearance/collapse.
Responsive Padding	<code>.section-padding</code> <code>py-12 md:py-20 lg:py-24</code>	Standardized vertical spacing; removes massive desktop padding (<code>py-32</code> / 128px) which was crushing mobile viewport space.
Touch Hover Override	<code>@media (hover: none) and (pointer: coarse)</code> <code>opacity: 1 !important</code>	Forces hover-dependent descriptions, technical badges, and CTA arrows to display permanently on touchscreen devices.
Custom Cursor Scoping	<code>@media (pointer: fine)</code>	Restricts custom SVG crosshair cursors strictly to desktop pointer devices, avoiding touch drag lag on phones and tablets.
Global Media Bounds	<code>img, video { max-width: 100%; height: auto; }</code>	Guarantees no SVG icon, image, or video element can physically exceed its container width and trigger horizontal scrolling.
iOS Safari Zoom Guard	<code>input, textarea { font-size: 16px; }</code>	Prevents Apple iOS Safari from automatically zooming into the webpage whenever a user taps an inquiry input field.
Horizontal Overflow	<code>body { overflow-x: hidden; }</code>	Prevents erratic side-to-side page wobbling on mobile touch devices.

2 Global Navigation & Mobile Header

src/app/App.tsx (Nav & MobileDrawer)

- **Extended Mobile Breakpoint:** Mobile navigation trigger extended to tablet screens (`<= 1024px` / `lg:hidden`) rather than just small phones, avoiding menu wrap collision.
- **Animated Hamburger Icon:** Replaced static icon with smooth CSS-animated SVG hamburger/close toggle.
- **Dedicated Mobile CTA:** Added a high-contrast tactical `Contact Us` button directly in the mobile sticky header bar.
- **Mobile Drawer UX:**
 - Configured background body scroll-lock (`overflow: hidden` on body) with automatic viewport resize cleanup.
 - Touch targets expanded to WCAG 2.1 compliance (minimum `44px` touch hit area).
 - Dynamic drawer top calculation: `top-[80px]` when scrolled vs `top-[86px]` at page top to prevent overlapping the sticky navbar.
 - Removed redundant "Home" link to maintain exact parity with the desktop navigation menu.

3 Hero Sections (Product, System & Domain Pages)

30+ files across `src/pages/*/components/HeroSection.tsx` and `*Hero.tsx`

- **Dynamic Viewport Sizing:** Upgraded from fixed `h-screen` to `min-h-[100dvh] landscape:min-h-[85vh]` . Prevents content clipping when phones are rotated horizontally.
- **Fluid Typography Scaling:** Clamped oversized headings (previously static `text-6xl` to `text-8xl`) down to: `text-3xl sm:text-4xl md:text-5xl lg:text-7xl font-bold tracking-tight`
- **Standardized Mobile CTA Alignment:** All CTA button clusters align to the left (`justify-start`) on mobile for readable hierarchy, while preserving centered/spaced layouts on desktop.
- **3D Canvas Hero Viewport Listener (Homepage):** `src/components/CanvasScrollHero.tsx` Hides the heavy multi-megabyte 3D canvas image-sequence on viewports `<= 820px` , seamlessly rendering the lightweight static hero for high-speed mobile performance.

4 Interactive 3D / 360° Product Viewers

`src/components/Interactive360Viewer.tsx` | `Interactive360RhinoBlackViewer.tsx` | `Interactive360SpearViewer.tsx`

- **Non-Blocking Page Scroll:** Applied `touch-action: pan-y` to the canvas container so users can swipe vertically past the 360 viewer without page scroll trapping.
- **Touch Affordance Cue:** Added an animated floating indicator (`"DRAG TO ROTATE 360°"`) visible specifically on touch devices.
- **Aspect Ratio Constraints:** Implemented `aspect-[4/3] sm:aspect-[16/9]` to prevent tall vertical stretching on mobile screens.

5 Interactive Blueprint & Platform Variants

`src/components/InteractiveBlueprint.tsx`

- **Mobile Accordion Conversion:** On screens `< 768px` , the complex desktop horizontal multi-tab matrix switches automatically to an expandable vertical accordion.
- **Horizontal Touch Pill Navigation:** Added `overflow-x-auto no-scrollbar` with touch snap for seamless category filtering.
- **Minimum Hit Targets:** Tab buttons formatted with minimum `48px` height for finger taps.

6 System Ecosystems & Horizontal Product Tracks

11+ files: `AerospaceDefenceEcosystem`, `DroneSystemsEcosystem`, `ElectroOptics`, `RadarSystems`, etc.

- **Fluid Card Widths:** Replaced rigid `w-[1100px]` with responsive units: `w-[88vw] sm:w-[75vw] lg:w-[60vw] max-w-[1100px]` .
- **Touch Inertia Swiping:** Added `-webkit-overflow-scrolling: touch` and `scroll-snap-type: x mandatory` .
- **GSAP MatchMedia Isolation:** Wrapped ScrollTrigger pinning inside `gsap.matchMedia('(min-width: 768px)')` . On mobile, pinning is cleanly bypassed to prevent jumpy scroll locking.

7 Use Cases & Operational Capabilities Grids

20+ files: `src/pages/*/components/UseCasesSection.tsx`

- **Clean Responsive Progression:** Corrected breakpoint overrides to ensure:
`grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 gap-6 sm:gap-8`
- Eliminated breakpoint conflicts that were forcing tablet viewports into single-column layouts.
- Added responsive card padding (`p-6 sm:p-8`) to maintain breathing room around text on small devices.

8 Contact & Business Inquiry Modal

src/components/ContactModal.tsx

- **Dual-Column to Single-Column Shift:** On screens `< 768px`, the marketing brand banner is hidden (`hidden md:block`) and the form takes `100%` modal width.
- **Keyboard Clearance:** Configured `max-h-[90dvh] overflow-y-auto overscroll-contain` so inputs and submit buttons stay accessible above the mobile software keyboard.
- **Direct Input Tap Targets:** Inputs styled with `min-h-[48px]` and minimum `16px` base font to eliminate iOS Safari auto-zoom.

9 Global Footer Section

src/app/App.tsx (Footer)

- **Grid Collapsing:** Reconfigured footer columns from fixed 4-column row into responsive `grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-8`.
- **Tap Target Spacing:** Added vertical touch padding (`py-2`) to all footer links (`min-h-[44px]` hit target).
- **Visual Consistency:** Standardized footer background to `bg-[#1a1a1a]` with high-contrast asset rendering.

10 Key Page-Specific QA Updates

Page	Component File(s)	Fix Applied
About Us	AboutManifesto.tsx AboutJourney.tsx AboutLeadership.tsx	Reduced excess top/bottom padding in Manifesto. Converted desktop horizontal timeline to responsive vertical milestone cards. Executive leadership cards collapse to <code>grid-cols-1 sm:grid-cols-2 lg:grid-cols-3</code> .
Clientele & Alliances	ClientDirectory.tsx ClientMetrics.tsx FeaturedAlliances.tsx	Client logo directory updated to <code>grid-cols-2 sm:grid-cols-3 md:grid-cols-4</code> . Key metric cards collapse to 2 columns on phones. Horizontal alliance cards enabled with hidden touch scrollbars.
Newsroom	NewsroomPage.tsx	Press release cards stacked vertically on mobile (<code>flex-col md:flex-row</code>). Featured article media ratio scaled to <code>aspect-[16/10]</code> .
Investor Relations	FinancialReportsPage.tsx GeneralMeetingNoticePage.tsx	Converted wide multi-column document tables into responsive card rows with full-width PDF download buttons for thumb reach.

11 Developer Deployment & Verification Checklist

- ✓ **Verify theme.css Inclusion:** Ensure your production stylesheet includes `.section-padding` , `min-h-[100dvh]` , and the `@media (hover: none)` rules.
- ✓ **Verify Bundle Assets:** Upload the compiled distribution bundle (from `dist.zip`) to your server's WordPress public frontend folder.
- ✓ **Clean Caches:** Flush WordPress page caching (WP Rocket / W3 Total Cache / LiteSpeed) and CDN caching (Cloudflare) so new CSS bundles serve immediately.
- ✓ **Mobile Device Testing:** Test on physical iPhone (Safari) and Android (Chrome) to verify dynamic viewport heights, touch navigation drawer, and 360 viewer drag affordances.